INCH-POUND

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MILITARY SPECIFICATION

VALVE, OXYGEN MASK, COMBINATION INHALATION AND EXHALATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This Specification covers a pressure-compensating inhalation valve, for use with oxygen mask assemblies MBU-5/P and MBU-12/P.

* 1.2 Classification. Valves shall be of the following types:

Type I - Use with MBU-5/P oxygen mask assembly.

Type II - Use with MBU-WP oxygen mask assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form apart of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-A-1034	Compressed Air, Breathing
BB-N-411	Nitrogen, Technical

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technology & Industrial Services Division, SA-ALC/TILDD, 485 Quentin Roosevelt Rd, Kelly AFB, TX 78241-6425 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116	Preservation Methods of
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-M-26542/3	Microphone, M-100/AIC
MIL-M-26542/12	Microphone, M-169/AIC
MIL-O-27210	Oxygen Aviator's Breathing Liquid and Gas
MIL-M-27274	Mask, Oxygen, MBU-5P
MIL-M-87163	Mask, Oxygen, MBU-12/P

STANDARDS

FEDERAL

FED-STD-595

(Requirements for individual color chips, 3 x 5 supplements should be submitted to the Stadardization Documents Order Desk Building 4D, 700 Robbins Avenue Philadelphia PA 19111-5094 citing FED-STD-595 together with appropriate chip number as shown therein.)

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-781	Reliability Testing for Engineering Development, Qualification and production
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Disimilar Metals
MIL-STD-970	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-2073-l	DOD Materiel procedures for Development and Application of Packaging Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government Publications. The following documents forma part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUE, INC.

ANSI B46.1	Surface Texture (Surface Roughness, Waviness and Lay).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

(Non-Government standards and other publications are normally available from the organization that prepare or distribute the documents. These documents also maybe available in or through libraries or other informational services.)

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2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The combination inhalation and exhalation oxygen mask valve furnished under this specification shall be products which are authorized by the qualified activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.3).

3.2 Selections of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. Materials shall be as specified herein. When materials are used which are not specifically designated they shall be of the best quality and entirely suitable for the purpose intended. Unless otherwise Specified by the procuring activity, all materials and all component parts used in the valve shall have been manufactured not more than 12 months prior to the date of delivery Of the valves.

3.3.1 Nonmetallic materials. Any nonmetalic material that is easily deteriorated or otherwise adversely affected by continued service use with oxygen shall not be used. Whenever diaphragms or flappers are used in the construction of the type I valve, they shall be made of a silicone rubber. Diaphragms, flappers and the sealing and isolation washers for the type II valves shall be made of a silicone-type rubber. All silicone-type rubber must be acceptable to the procuring activity.

3.3.2 Dissimilar metals. Unless suitable protected against electrolytic corrosion dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Aluminum or aluminum allows. Aluminum or aluminum alloys,when used shall be anodized black in accordance with MIL-A-8625, type II

3.3.4 Protective treatment. When materials are used in the construction of the valve that are subject to corrosion in oxygen, salt air, or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective mating that will crack, chip, or scale with age of extremes of atmospheric conditions shall be avoided.

3.3.5 Corrosion-resistant. All metals parts shall be of the corrosion-resistant type or suitably treated to resist corrosion. Brass, bronze, phosphor bronze, 18-8 corrosion-resistant steel, and anodized aluminum alloy are considered satisfactory corrosion-resistant materials.

3.3.6 Silicone rubber. All silicone rubber used in the valves shall be smooth and free from flash, foreign materials, imperfections, and objectionable odors.

3.3.7 Toxicity. The material used in the valves shall not contain any chemicals or materials that have toxic properties as a result of being cleaned with 70-percent isopropyl alcohol solution.

3.4 Desire. The valves shall be as defined by figure 1 and shall conform to the envelope dimensions shown in figure 2. They shall each consist of a valve body, a cap conforming to figure 3, and bearing and sealing washers as described for each type valve in figure 4 and 5. The valves shall be designed to control the oxygen pressure within a mask and to respond automatically to changes in the regulator. The valves shall be mounted in the expiatory chamber of pressure breathing type masks and shall form a leak tight fit in the exhalation port. Additionally, care shall be taken in the type I valve design to assure that the exhalation ports on the valve shall not become blocked by the inner walls of the mask's rubber anti-icing boot. The valves shall embody the design

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principles of pressure compenstion and damping. The inhalation-exhalation device of figure 2 shall be considered as a typical method of design Any deviations in envelope or critical interface dimensions from figures 2,3,4 and 5 may, at the option of the procuring activity, require the contractor to install the valve in all sizes of the appropriate type oxygen mask assembly as described in MIL-M-27274 and MIL-M-87163. The contractor will then be required to demonstrate to the procuring activity, the ability of the valve to meet the requirements of the following performance tests as per MIL-M-27274 and MIL-M-87163:

Mask Leakage
Combination Valve Installation Inlet Performance
Combination Valve Installation, Exhalation performance
Combination Valve Installation, Exhalation Resistance

3.4.1 Valve sticking. The exhalation valves may have a tendency to stick in the closed position if there is a sufficient buildup of sticky residue on the exhalation valve seat, the valves shall be so designed that the valve seat can be cleaned by first removing the valve cap and then swabbing the valve seat with a cotton-tipped swab saturated with a warm diluted solution of a mild detergent and water followed by a rinse with distilled water.

3.4.2 Inadvertent oxygen regulator shut-off protection - The design of the valves shall include a feature which shall provide protection during inadvertent oxygen regulator shut-off. This feature shall prevent the breathing of ambient air through the valve.

* 3.4.3 Reliability. The valves shall have specified mean cycles between failures (MCBF) of not less than 180,000 cycles at the 90 percent lower confidence limit (see 4.6.22). (This equivalent to a minimum acceptable MCBF of 120,000 cycles and a reliability of 0.9775 for a 10-hour, 9,000 cycle mission.)

3.5 Construction. The valves shall withstand the strains, jars, vibration, and other conditions incident to shipping,storage,installation, and service. There shall be no undue flutter or vibration of the valves under service conditions. The valves shall be so constructed that they cannot be disassembled without visible evidence, since it is intended to replace the entire valve instead of permitting repairs when failure occurs. If threads are used in the assembly of the body of the valves, the threads shall be sealed by a method approved by the procuring activity.

3.6 Performance.

3.6.1 Dumping. The differential pressure shall not fall below 0.1 inch of water (gauge) with a regulator pressure of 0.5 to 14 inches of water (gauge) and an increase of flow from 1.0 to 150 liters per minute (see 4.6.2).

3.6.2 Leakage - suction. Leakage of the valves shall not exceed 0.1 liter per minute when the valve is subjected to the leakage-suction test (4.6.3).

3.6.3 Leakage - suction, low inlet valve. Back leakage through the inlet check valve shall be not more than 15 milliliters per minute with a suction of 0.1 inch of water (gauge) (see 4.6.4).

3.6.4 Inhalation resistance. The inhalation resistance of the valve shall not exceed 0.75 inch of water (gauge) at a flow of 50 liters per minute and shall not exceed 1.50 inches of water (gauge) at a flow of 100 liters per minute (see 4.6.5).

3.6.5 Leakage - pressure. Leakage shall not exceed 0.1 liter per minute with a flow of 1.0 to 50 liters per minute at a pressure of 0.5 to 14 inches of water (gauge) (4.6.6).

l 3.6.6 Exhalation resistance. Differential pressure shall be between 0.1 and 1.5 inches of water (gauge) with flows in the range of 0.1 to 50 liters per minute (see 4.6.7).

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* 3.6.7 Exhalation resistance - pressure breathing. The differential pressure shall be between 0.1 and 1.5 inches of water (gage) with a flow in the range of 0.1 and 50 liters per minute and a pressure within the range of 0.5 to 14 inches of water (gauge) (see 4.6.8).

* 3.6.8 Overload. The Valve shall not leak in excess of 1.0 liter per minute when a pressure of 20 inches of water (gauge) is applied to both sides of the valve (4.6.9).

* 3.6.9 Capacity. The differential pressure shall not exceed 2.6 inches of water (gauge) with a regulator pressure of 0.5 to 14 inches of water (gauge) and a water flow increasing through the ranged 10 to 100 liters per minute (see 4.6.10).

* 3.6.10 Leakage - suction, high inlet valve. Back leakage through the valve shall be not more than 150 mililiters per minute with a suction of 20 inches of water (gauge) (see 4.6.11).

3.6.11 High and low temperature operation. The valve shall be capable of satisfactory operation at temperature of +160°F and -40°F (see 4.6.12).

* 3.6.12 Burst pressure. The valve diaphragm shall be capable of withstanding a pressure of 5 PSIG without rupture (see 4.6.13).

3.6.13 Low temperature storage and vibration. The valve shall be capable of satisfactory operation after vibration at -65°F and exposure to -85°F (see 4.6.14).

3.6.14 Orientation. The valve shall be capable of satisfactory operation with the reference axis of the valve in the horizontal position (see 4.6.15).

3.6.15 Effects on communication. The valve shall not adversely affect speech communication at any altitude from zero to 32,000 feet (see 4.6.16).

3.6.16 Acceleration unseating of exhalation valve. The exhalation valve shall not be reseated when the combination valve is subjected to accelerations up to nine times the force of gravity (see 4.6.17).

3.6.17 Suction unseating of exhalaton valve. Back leakage through the exhalation valve shall be not more than 100 milliliters per minute when a suction of 10 inches of water (gauge) at the regulator end and a suction of 5 inches Of Water (gauge) at the mask end of the valve are applied simultaneously (see 4.6.18).

3.6.18 Subjective use. When insalled in a type MBU-5/P oxygen mask conforming to MIL-M-27274 or an MBU-12/P oxygen mask conforming to MIL-M-87163, as applicable, and used by human subjects, the valve shall not have any detrimental or objectionable performance characteristics at altitudes from ground level to 50,000 feet at temperatures from -40°F to +118°F, or any combinations of such altitude and temperature. The valve shall also operate at 3,4,5,6,7,8 and 9 times the force of gravity when the subject is performing an anti-g straining maneuver (see 4.6.19).

* 3.6.19 Restraint cable retention pin strength. The valve shall have a restraint cable retention pin similar to the one shown on figure 2. The valve body and cap shall not be damaged or pulled free, and the restraint cable retention pin shall not be pulled free from its anchor point when tested as specified (see 4.6.20).

* 3.6.20 Valve installation strength. This test only applies to the MBU-12/P oxygen mask. The valve body and cap shall not fail or be separated when tested as specified (see 4.6.21).

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* 3.6.2.1 Exhalation mechanism leakage. The valve shall be designed such that no portionof the exhalation mechanism will stick in the open position and allow leakage of the ambient air into the breathing air. This is of most concern when the valve is an a chilled environmenta and is then placed in a warm moist environment as tested in 4.6.23. After stowage at a temperature of 50 to 55°F followed by immediate subjective use, valve shall meet the requirements specified (see 3.6.10).

3.7 Interchangability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern changes in the rnanufacturer's part numbers.

3.8 Weight The weight of the valve including the valve body, the cap and bearing, sealing or isolation washers, as applicable, shall not exceed 35 grams.

3.9 Color. The exterior of the valve body shall conform to color No. 37038 for FED-STD-595.

3.10 Identification of product. A nameplate, permanently and legibly marked in accordance with MIL-STD-130, shall be securely attached or stamped in a conspicuous place on the valve, but shall not interfere with the operation of the valve.

3.11 Workmanship. Inasmuch as the operation of this valve directly affects the safety of personnel, workmanship shall be equal to or better than the highest grade of aircraft instrument manufacturing practice to ensure minimum opportunity for the development of defects in service. The valve shall be clean and free from any defect that might affect functionality. It shall be smooth and free of sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsible for Inspection. Unless otherwise specified in the contractor purchase order, the contractor is responsive for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contractor purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to peform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspections as part of manufacturing operations, is an acceptable practice to ascertain Conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Test conditions.

4.3.1 Reference conditions. Valves, rotometers, pressure and vacuum sources, and water columns shall be in accordance with figure 6. Unless otherwise specified, testing shall be conducted with the reference axis in a vertical-position.

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* 4.3.2 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77°F to ± 18°F, and a relative humidity of 60 t0 80 percent. Where tests are made with an atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.3.3 Gas. The high pressure gas used in testing the valves shall be oxygen conforming to type I of MIL-O-27210; water-pumped nitrogen conforming to type I, class I, grade B of BB-N-411; or compressed air conforming to Federal Specification BB-A-1034, compressed air breathing, source II, Grade B.The water content shall be 0.3 mg per liter maximum. If nitrogen or air is used, appropriate density correction factors shall be applied to the rotometer used, not only to correct the effect on the meter, but also on the performance of the valve with lower density gas.

4.3.4 Test stand. The tests specified (see 4.6.2,4.6.3,4.6.6,4.6.7, 4.6.8,4.6.9, and 4.6.15) shall be conducted by means of a test stand conforming to figure 6. The other tests shall be conducted by means of any suitable apparatus. The equipment for the test stand shall conform to table I. As an alternate, test equipment capable of performing the identicle test specified herein may be used when specifically approved by the procuring activity. All tests of valves shall be accomplished on the test stand with a gasket that duplicates the exhalation port and anti-king chamber of the MBU-5/P oxygen mask conforming to MIL-M-27274 or the hardshell exhalation chamber of an MBU-12/P mask conforming to MIL-M-87163, as applicable.

TABLE I Test stand equipment

Equipment	Description
Valve No	
1 and 2	1-inch internal pipe thread, quick-acting gate valves.
*3, 4, 5, 7 & 8	Needle valves - any size.
6	3/4-inch internal pipe thread, 2-way plug.
Rotometer No	
1	0 to 150 liters per minute at 14.7 PSI
2	0 to 2.5 liters per minute at 14.7 PSI
3	0 to 2.5 liters per minute at 14.7 PSI
Water Column	
A	30-inch (min) water column. Indicates regulator pressure from source "A."
B	30-inch (min) water column. Indicates differential pressure between pressure source "B" and pressure source "A."

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* 4.3.4.1 Pressure regulation. Pressure sources A and B, as shown on figure 6, shall be capable of being regulated from 0 to at least 20 inches of water pressure (gauge). Pressure source B shall be capable of being regulated between 0 to at least 14 inches of water pressure (gauge) with a maximum flow of 100 liters per minute. The vacuum source shall be capable of being regulated within the range of 0 to 15 inches of water (gauge) vacuum at a flow of 1 liter per minute.

4.3.4.2 Valve orintation for test. Unless otherwise specified, all valve tests shall be conducted with the axis of the valve oriented so that the form of gravity shall tend to assist the exhalation valve in opening.

4.3.5 procedure for acceptable test In conducting these&s@ at least two points within the ranges specified in the applicable test shall be obtained- One of these points shall include the maximum pressure or vacuum as applicable, and the maximum flow, when applicable.

4.4 Qualification inspection.

4.4.1 Qualification samples. The qualification samples shall consist of two valves with the exception Of the restraint cable retention pin and valve cap installation strength tests which shall consist of a sample of 20 valves. Samples shall be identified with the manufacturer's own part number.

4.4.2 Test samples and test reports for the qualifying activity. The qualification test samples and three copies of a qualification test report prepared in accordance with MIL-STD-831 shall be finished to the qualifying activity.

4.4.3 Qualification inspection. The qualification inspection shall consist of all the tests described under 4.6.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of:

a. Individual tests.

b. Sampling tests.

4.5.1 Individual tests. Each valve shall be subjected to the following tests as described under 4.6:

a. Examination of product.

b. Dumping.

c. Leakage - suction

d. Leakage suction low inlet valve.

e. Inhalation resistance.

f. Leakage - pressure.

g. Exhalation resistance.

h. Exhalation resistance - pressure breathing.

4.5.2 Sampling tests. Five valves manufactured under essentially the same conditions and submitted for acceptance at substamtially the same time shall be selected at random from each lot of 500 valves, or fraction thereof, and subjected to the following tests as described under 4.6:

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a. overload.

b. Capacity

c. Leakage - suction high inlet valve.

d. Burst pressure.

e. Reliability test.

f. Suction unseating of exhalation valve.

g. Restraint cable retention pin strength.

4.5.2.1 Rejection and retest When one or more items from a lot fail to meet the specification, acceoptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor Shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests maybe continued pending the investigation of a sampling test failure; but final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may

to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. Each valve sha1l be carefully examined to determine conformance to the requirements of this specification with respect to materials workmanship, design construction, and marking.

* 4.6.2 Dumping. With a regulator pressure of 0.5 to 14 inches of water (gauge), the flow shall be increased from 1.0 to 150 liters per minute. The differential pressure shall not fall below 0.1 inch of water (gauge) (mask pressure exceeding regulator pressure) at any time during this test.

4.6.3 Leakage - suction. The following test shall be performed: Close valves 1,2,3 and 8 and open valves 4, 5, and 7 shown on figure 6. Adjust the vacuum source within the range of O.3 to 15 inches on water (gauge) vacuum as shorn on water column A. The leakage, read on rotometer 3, shall not exceed 0.1 liter per minute.

4.5.4 Leakage- suction, low inlet valve. A suction of 0.1 inch of water (gauge) shall be applied to the hose attach port side of the valve. Back leakage through the inlet check valve shall be not more than 15 milliliters per minute.

4.6.5 Inhalation resistance. The inhalation resistance of the valve shall not exceed 0.75 inch of water (gauge) when exposed to a flow of 50 liters per minute and shall not exceed 1.50 inches of water (gauge) when exposed to a flow of 100 liters per minute.

* 4.6.6 Leakage - pressure. The following test shall be performed: Open valves 1,2,3 and 7 and close valves 4, 5, and 8 shown on figure 6. Regulate pressure source B to obtian flow in the range of l.0 to 40 liters per minute, read on the rotometers 1 and 2. Set pressure source A within the range of 0.5 to 14 inches of water (gaige), read on water column A. Reduce pressure source B in order that a reading of 0 is obtained on water column A. Reduce pressure source B in order that a reading of 0 is obtained on water column B. The reading obtained on rotometer 2 shall not exceed 0.1 liter per minute.

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4.6.7 Exhalation resistance. The following test shall be performed: Open wolves 1, 2, and 8 and close valves 3,4,5 and 7 shown on figure 6. Vary the pressure source B in order that flows in the range of O.l to 50 liters per minute may be observed on rotometers 1 and 2. The differential pressure, read on water column B, shall be between 0.1 and 1.5 inches of water (gauge). However, over the flow range of 0.1 to 1.0 liters per minute, water column B shall not exceed 0.70 inches of water (gauge).

* 4.6.8 Exhalation resistance - pressure breathing. The following test shall be performed: Open valves 1,2,3 and 8 and close valves 4, 5 and 7 shown on figure 6. Regulate pressure source B to obtain flows in the range of 0.1 to 50 titers per minute, read on rotometer 1 or 2. Set pressure source A within the range of 0.5 and 14 inches of water pressure (gauge), read on water column A. The differential pressure read on water column B shall be between 0.1 and 1.5 inches of water (gauge).

4.6.9 Overload. A pressure of 20 inches of water (gauge) shall be applied for a period of 2 minutes to both the hose attach port and mask sides of the valve. The valve shall not leak in excess of 1.0 liter per minute under these conditions.

4.6.10 Capacity. With a regulator pressure of 0.5 to 14 inches of water (gauge), the flow shall be increased through the range of 10 to lOO 1iters per minute. The differential pressure indicated on water column B shall not exceed 2.6 inches of water (gauge).

4.6.11 Leakage - suction high inlet valve. A suction of 20 inches of water (gauge) shall be applied to the hose attach port side of the valve. Back leakage through the valve shall be not more than 150 milliliters per minute.

4.6.12 High and low temperature operation. The valve shall be exposed to temperature of +160°F and -40°F for 4 hours at each temperature. After each exposure and while still at the test temperate, the valve shall be tested as specified (see 4.6.6, 4.6.7 and 4.6.8). For the low tempemture exposure, the flow rates shall be in the range of 4.0 to 50 liters per minute instead of 0.1 to 50 liters per minute.

4.6.13 Burst pressure. A pressure source which can be regulated within the range of 0 to 10 PSIG shall be connected to the lower protuberance of the valve. In the hose, between the regulated source and the valve, insert a tee connection to which is connected a manometer or gauge, calibrated in pounds per square inch Rupture shall not cccur at less than 5 PSIG.

4.6.14 Low temperature storage and vibration. The valve shall be subjected to a temperature of -65°F ± 5°F and to vibration at this temperature for a period of not less than 3 hours. The vibration shall be between 500 and 3,000 cycles per minute and shall be of such an amplitude that a point of the valve will describe, in a plane inclined 45° to the horizontal, a circle of not less than 0.018 inch nor more than 0.020 inch diameter. After the 3-hour period the vibration shall be discontined and the valve shall be further cooled to a temperature of -85°F ± 2°F. After 48 hours at this temperature, the valve shall be allowed to reach room temperature and then shall be subjected to the test specified (see 4.6.9).

4.6.15 Orientation. The test specified (see 4.6.3, 4.6.6, 4.6.7, and 4.6.8) shall be repeated with the reference axis of the valve in a horizontal position. The differential pressure indicated on water column B shall not increase by more than 0.5 inch of water (gauge) over the values previously measured at each point.

4.6.16 Effects on communication. Subjective tests shall be performed on the valve to determine effects of valve operation in regard to communications. The valve to be tested shall be installed in an MBU-5/P pressure-breathing mask conforming to MIL-M-27274 or an MBU-12/P pressure-breathing mask conforming to MIL-M-87163, as applicable, and shall be equipped with a type M-100/AIC or M-169/AIC microphone, as applicable, conforming to MIL-M-26542/3 or/12. This microphone shall be connected to a Type AN/AIC-18

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intercommunication set. The effects on communications shall be determined by listening qualitatively to speech communication from at least three talkers when using the valve at any altitude between ground level and 32,000 feet. The exhalation valve shall not whistle, chatter, or create noise excessively to a degree which interferes with communication. The exhalation valve shall not modulate the speech signal because of valve flutter.

* 4.6.17 Acceleration unseating of exhalation valve. The combination valve shall be mounted on a horizontal centrifuge so that axial acceleration is applied from the mask end to the regulator end of the valve. The valve shall then be subjected to acceleration from 0 to 9 times the force of gravity. The exhalation valve shall not be unseated at any acceleration up to nine times the force of gravity.

4.6.18 Suction unseating of exhalation valve. A suction of 10 inches of water (gauge) shall be applied to the regulator end of the valveand held. A suction of 5 inches of water (gauge) shall be applied to the mask end of the valve. Leakage through the exhalation valve shall be measured. A maximum leakage of 100 milliliters per minute will be permissible during this test.

4.6.19 Subjective use. The valve shall be Wed in a type MBU-5/P or MBU-12/P oxygen mask as applicable, and used by human subjects at any altitude, temperature, or combination altitude-temperature as specified in 3.6.18 for periods up to 2 hours. The valve shall also be placed in an MBU-5/P oxygen mask conforming to MIL-M-27274 or MBU-12/P oxygen mask conforming to MIL-M-87163, as applicable, and used by human subjects at a temperature of 0°F for a minimum of 2 hours. This is to insure that both during and after the test there is no freezing of the valve. High G tests shall be performed with at least 5th and 95th percentile subjects to determine that the valve will perform Satisfactorily at 3, 4, 5, 6, 7, 8 and 9 times the force of gravity. High altitude decompression tests which simulate aircraft decompression shall be performed to insure that the subject is able to breathe both during and after decompression. Subjective use tests are optional tests to be conducted at the discretion of and by the procuring activity when the valve appears to have performance characteristics that might be detrimental or objectionable for human use.

4.6.20 Restraint cable retention pin strength. The valve shall be held in place by the cap. A 100 pound load shall be applied to the restraint cable retention pin at its mid-section perpendicular to its longitudinal axis. The had shall be applied for 3 seconds.

4.6.21 Valve,installation strength. This test only applies to the MBU-12/P oxygen mask. The valve shall be installed in a test rig with the bearing washer, both gaskets and flange representative of the nominal thickness of the hardshell of the MBU-12/P oxygen mask torqued to the nominal torque value as recommended by the valve manufacter, and set at rim angle of 30 degrees of the longitudinal axis of the valve to the pull angle. Each of 20 valves shall be pulled at a pull speed of 0.30 inches per second until failure or seperation of the valve from the test rig. The test rig may grip the valve at the surface of the valve where the breathing hose is clamped in place rather than by being pulled from the restraint cable and retention pin as this may fail before the valve body and cap separate. Should any of the 20 valves separate with a force of less than 100 pounds, this shall be be cause for failure of all valves.

* 4.6.22 Reliability test The reliability test shall consist of simulated breathing cycles with oxygen or air as the test fluid each cycle of the valve shall consist of an inlet breathing flow rate varying from 0 to 50 liters per minute and back to 0 liters per minute again. The cyclic rate shall be 25 cycles per minute. At least 50 percent of the reliability testing of the valve shall be conducted under pressure breathing editions at 3.0 inches -4.0 inches of water (gauge), the remainder being under demand breathing conditions. A failure shall be identified as any malfunction which could compromise the performance of the valve. The valve shall be tested every 24 hours as follows:

a. The leakage-Pressure test specified (see 4.6.6) shall be accomplished.

b. The pressure demand operation test specified (see 4.6.8) shall be accomplished.

A test stand conforming to figure 6 need not be used for failure determinations during reliability testing.

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* 4.6.23 Exhalation mechanism leakage. Four MBU-12/P oxygen mask assemblies each containing a properly installed valve will be chilled in a temperature of 50° to 55°F for at least one hour.Immediately upon removing the mask assemblies from the cooling chamber, individuals will breathe though each of these masks for 3 to 5 minutes with the regulator lever in the normal position the ambient air at 70°F to 80°F and a relative humidity of 70 to 90 percen. Upon completion of the breathing cycle, the valve will be examined to determine if any of the mechanism (such as the silicone rubber skirt sections) are stuck to any portion of the valve (such as the surrounding metal ring) or are otherwise stuck open. Any valve mechanism found stuck in the improper position for inhalation through the valve shall be cause for rejection of the valve. Next the valve shall be tested for back leakage as specified (see 4.6.11). Any back leakage of ambient air into the valve in excess of that allowed by this specification shall also because for rejection of the valve.

4.6.24 Preservation and packaging. Unless otherwise specified in the purchace order or contract, the supplier is responsible for the performance of all inspections required as listed in table III, MIL-P-116 for the method of preservation being accomplished.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified by the procuring activity, the valves shall be packaged for delivery in accordance with MIL-STD-2073-1.

5.1.1 Levels A and B. Each valve shall be preserved in accordance with method IA8 of MIL-P-116.

5.1.2 Level C. Each valve shall be provided preservation which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. The required level of packing, A, B, or C, shall be specified by the procuring activity (see 6.2).

5.2.1 Level A. All valves preserved as specified (see 5.1.1) shall be packaged in exterior-type, weather-resistant containers. All containers shall be closed, sealed, and reinforced in accordance with appendix thereto,

5.2.2 Level B. All valves preserved as specified (see 5-1.1) shall be packaged in domestic-type exterior containers. All containers shall be closed and stropped in accordance with appendix thereto.

5.2.3 Level C. All packaged valves which require over-packaging for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation a the lowest rate to the point of delivery. Containers shall meet the uniform freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking. The exterior and interior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. The following precautionary marking shall appear on each unit package.

"CAUTION
DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT OXYGEN VALVES."

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6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful, but is not mandatory.)

6.1 Intended use. The valve covered by this specification is intended for use in connection with a pressure demand oxygen regulator, or any other regulator which will supply oxygen to the user in such a manner a at altitudes above approximately 28,000 feet, a higher intrathoracic pressure than that of the ambient air is continually maintained. It is intended to be inserted into the expiratory chamber of an oxygen mask. A light check valve is incorporated within the valve. This check valve opens during inhalation, thus allowing the pressure in the inlet tubing and facepiece of the mask to equalize. Upon exhalation the check valve closes, and the slight excess of pressure in the facepiece over the inlet line pressure causes the exhalation valve to open.

6.2 Acquisition retirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Type of valve (see 1.2)

d. Qualification (see 3.1, 4.3, and 6.3).

f. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

g. Precautionary marking (see 5.3.1)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, QPL-27296, whether or not such products have actually been so listed by that date. The attention of the contractor is dial to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Government tested for qualification in order that they maybe eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Logistics Center (SA-ALC/TILDD), Kelly Air Force Base, Texas 78241-6425, and information pertaining to qualification Of products may be obtained from that activity.

6.3.1 Drawings. When requested. the manufacturer should submit engineering drawings and inspection reports in accordance with SD-6 (Provisions Governing Qualification).

6.4 Part or Identifying Number (PIN). The PIN is as follows:

M 27296-1 Example of PIN: M27296-1

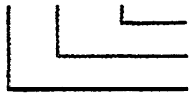


TABLE II Dash number and item

Dash No	Item
1	Type I valve
2	Type II valve

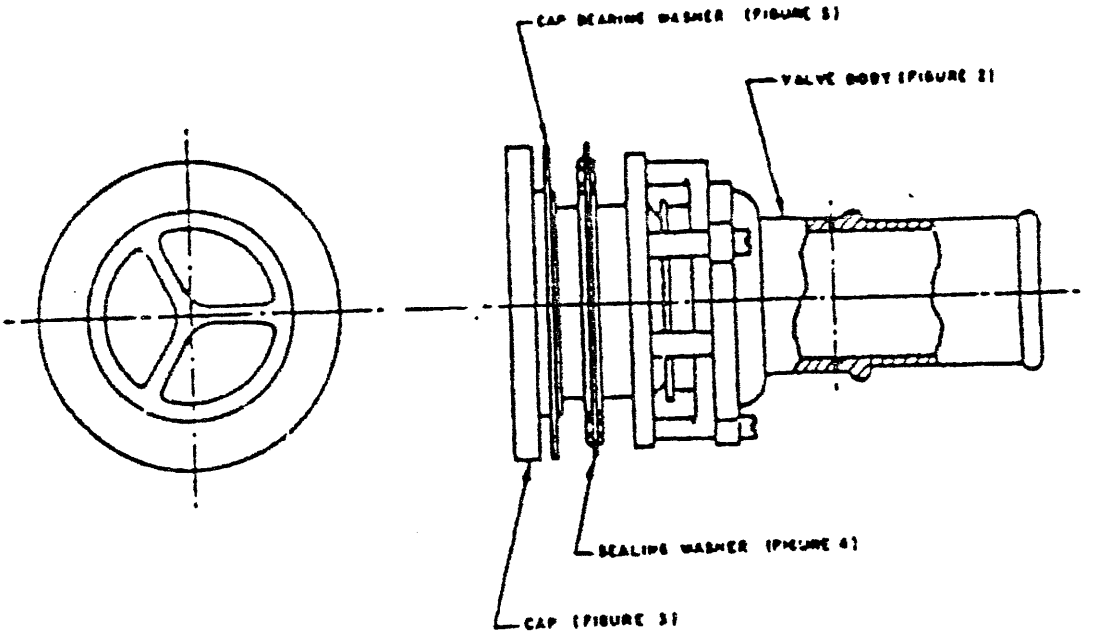
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6.5 Subject term (key word) listing.

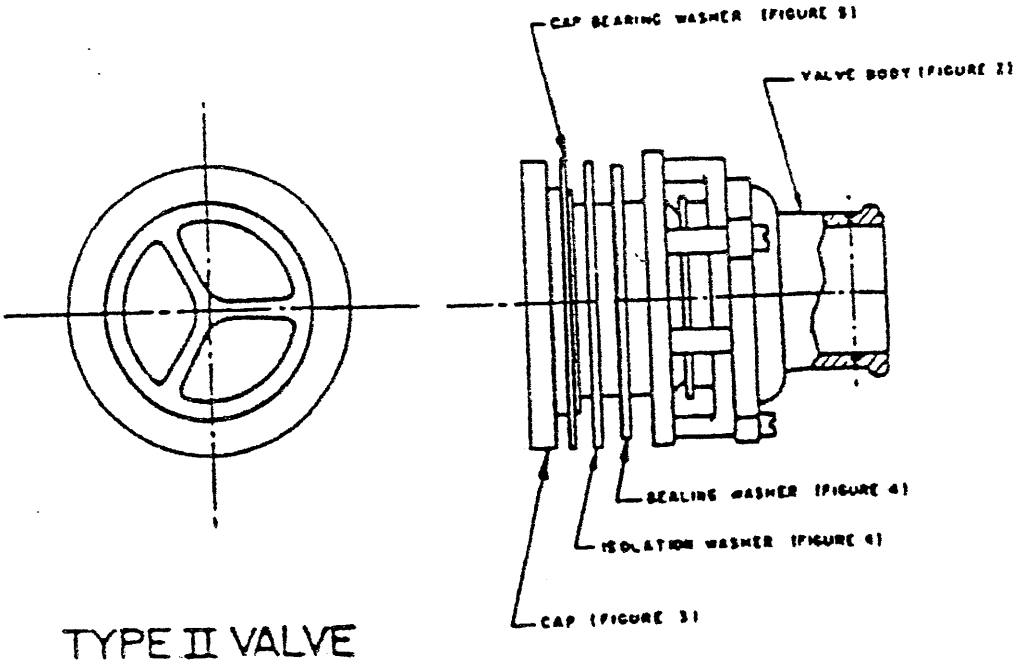
Cap
Communication
Leakage
Regulator
Temperature

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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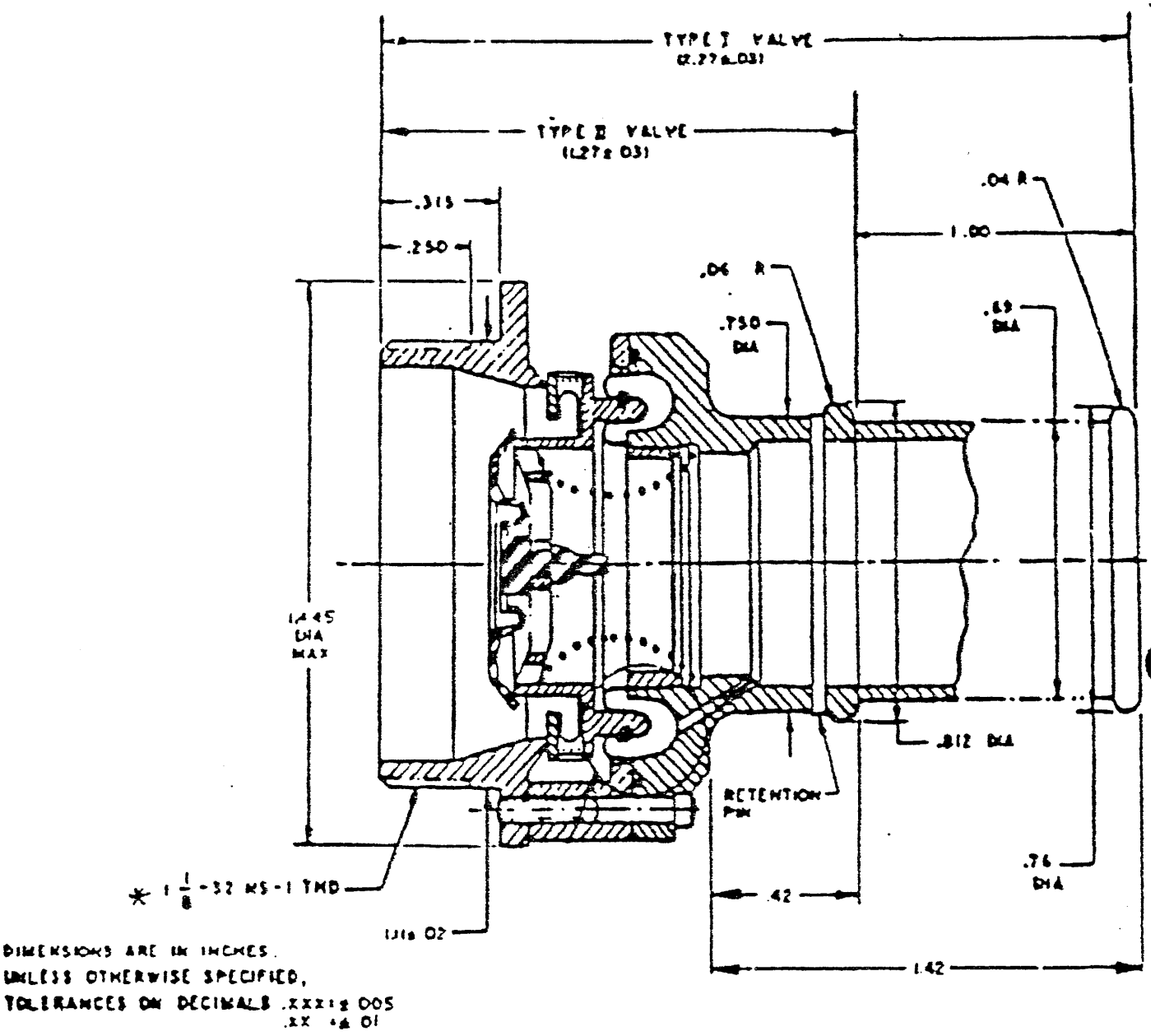
TYPE I VALVE



TYPE II VALVE

FIGURE 1. Washers and vale bodies

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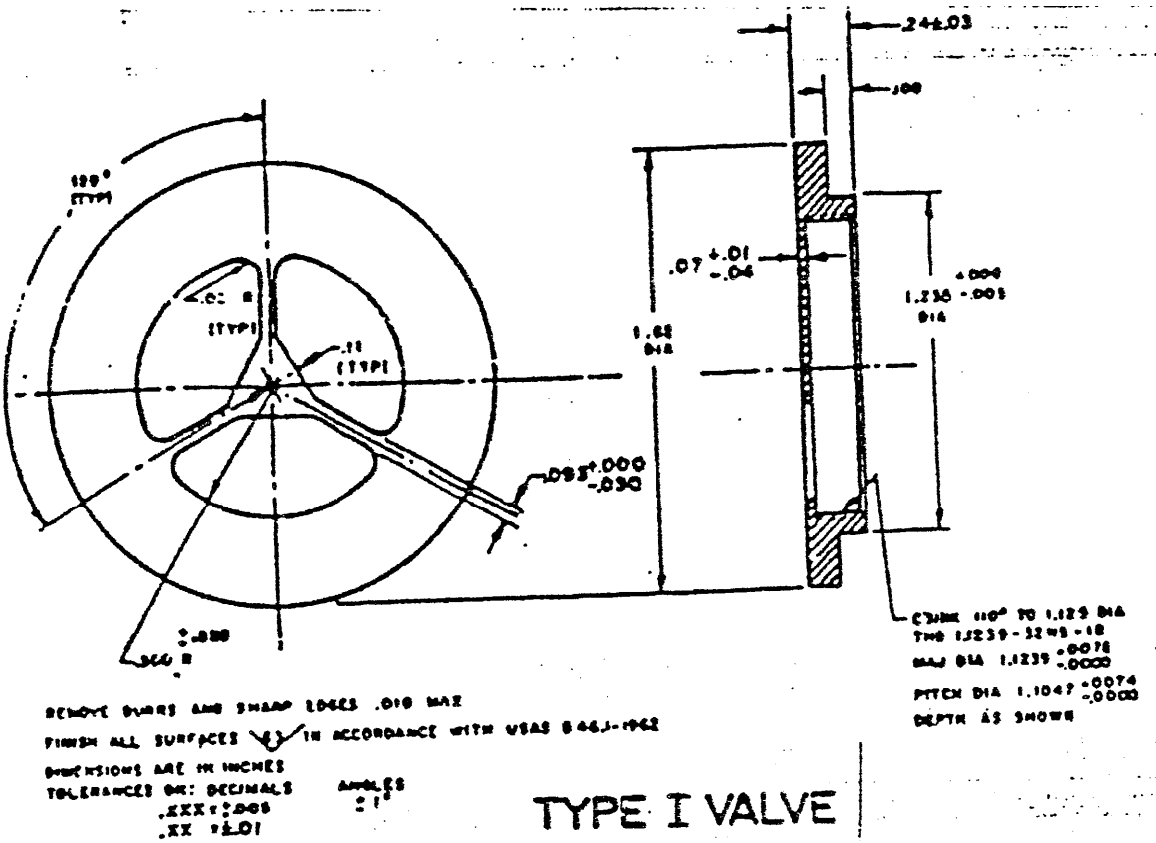


* NOTE: Threads may vary dependant on the valve design.

Nondimensional information is for illustrative purposes only.

FIGURE 2. Internal configuration (typical)

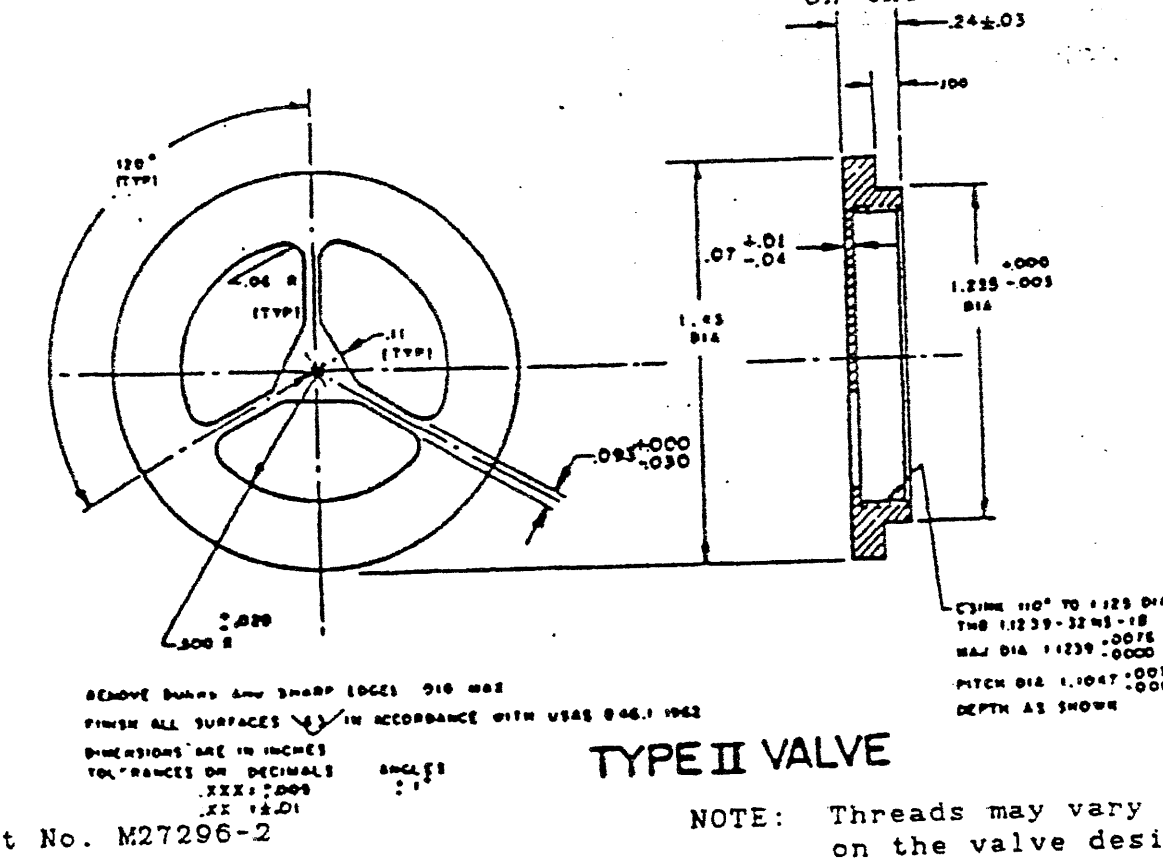
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TYPE I VALVE

Part No. M27296-1

NOTE: Threads may vary dependant on the valve design.



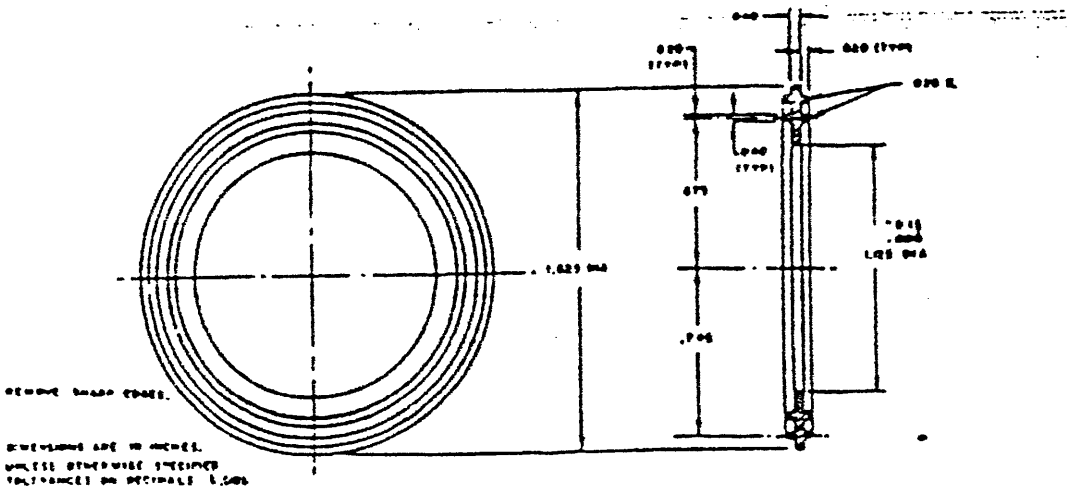
TYPE II VALVE

Part No. M27296-2

NOTE: Threads may vary dependant on the valve design.

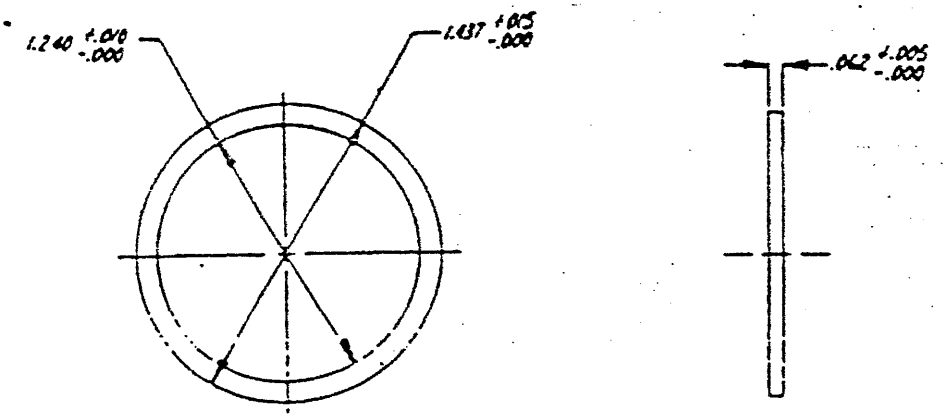
FIGURE 3. Caps

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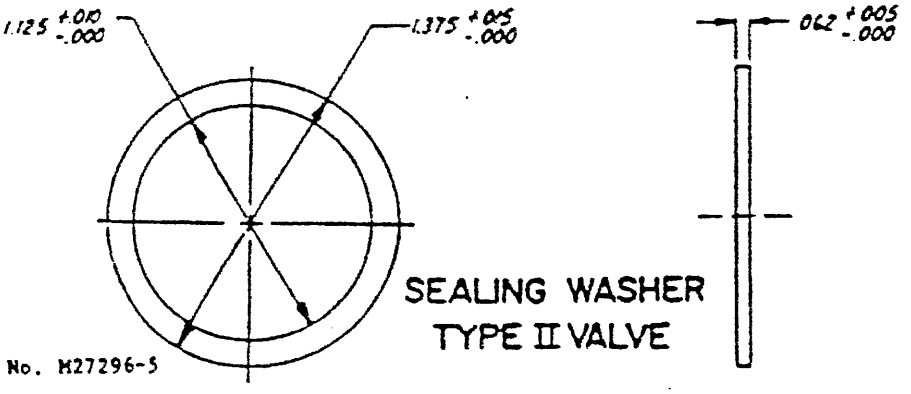
Part No. M27296-3

SEALING WASHER TYPE I VALVE



Part No. M27296-4

ISOLATION WASHER TYPE II VALVE

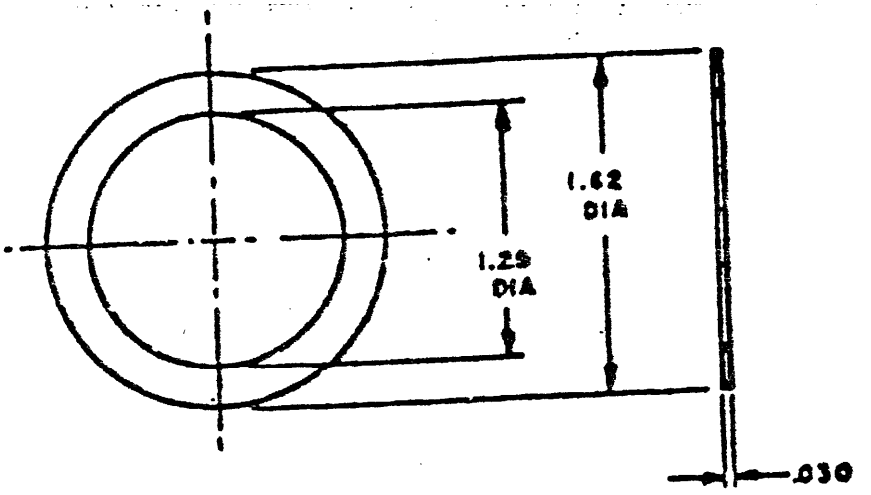


Part No. M27296-5

SEALING WASHER
TYPE II VALVE

FIGURE 4. Sealing and isolation washers

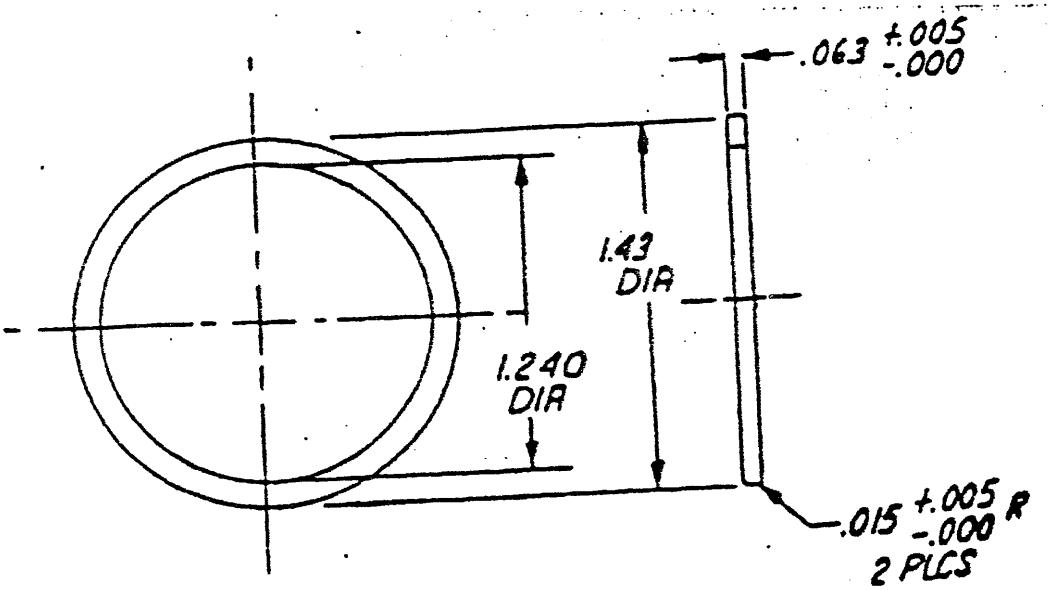
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DIMENSIONS ARE IN INCHES.
TOLERANCES ON DECIMALS .XXX=±.005
.XX =±.01

Part No. M27296-6

TYPE I VALVE



TYPE II VALVE

DIMENSIONS ARE IN INCHES,
TOLERANCES ON DECIMALS .XXX=±.005
.XX =±.01

Part No. M27296-7

FIGURE 5. Cap bearing washers

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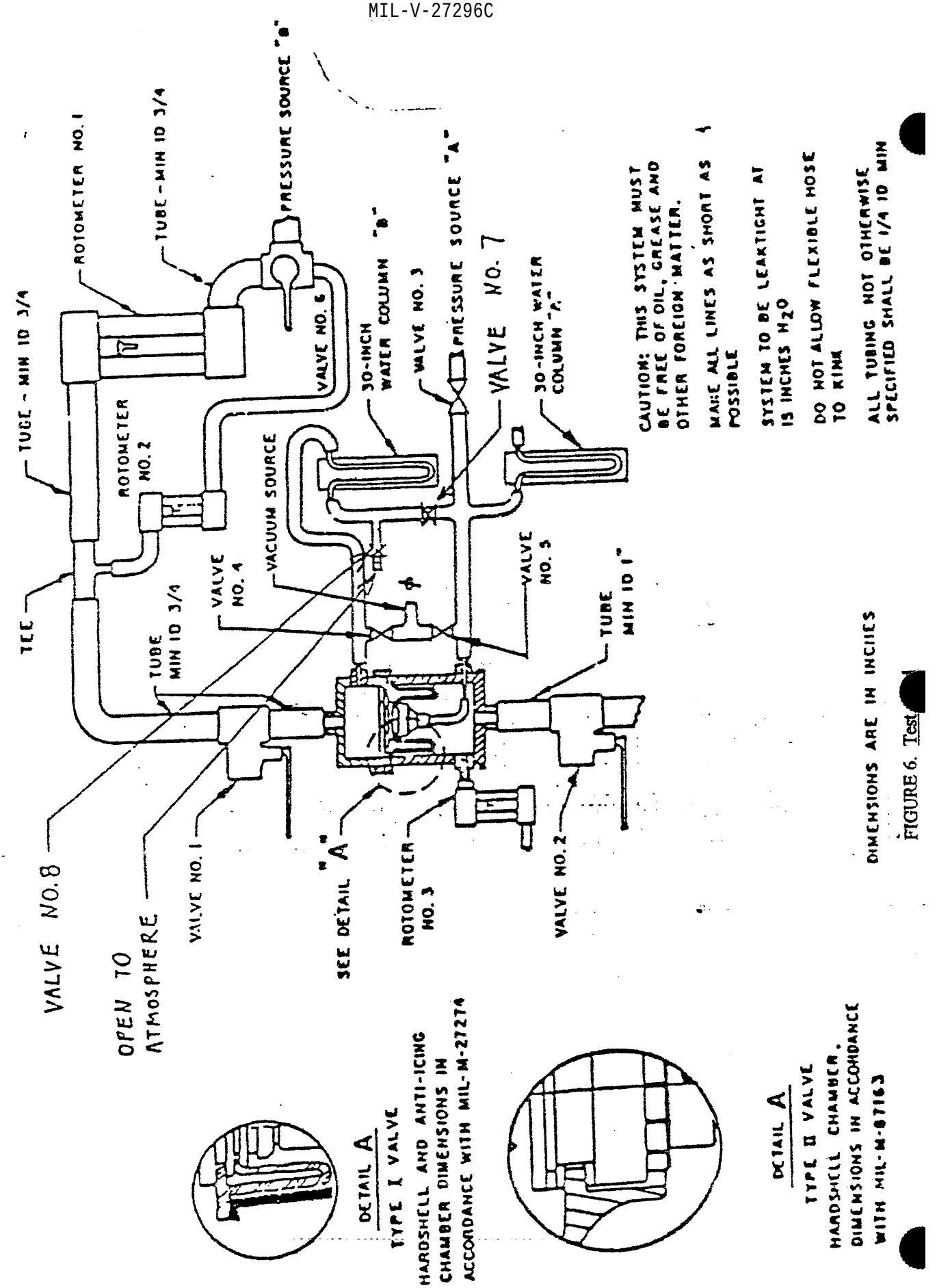


FIGURE 6. Test

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	X	X

3. DOCUMENT TITLE

X

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
X	
c. ADDRESS (Include Zip Code) X	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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